

BPC_01_VR_Locomotion Component

1. Component Overview

Purpose

BPC_01_VR_Locomotion is a VR character locomotion component attached to the VR character Blueprint. It extends and switches the VR character's movement methods. The component keeps the original teleportation feature and dynamically switches input mappings through a locomotion-type enum, toggling between teleport and smooth locomotion.

Design

- The enum variable VRLocationMode controls the current movement method (teleport / smooth locomotion)
- Input Mapping Contexts (IMCs) manage the input mapping for each movement method
- When UpdateMovementMethod is called, the array is iterated: mappings matching the current enum value are added, non-matching ones are removed
- In smooth locomotion mode, forward/back and left/right movement are based on the VR camera direction, so movement always matches where the player is facing
- The capsule collision follows the camera position to prevent the camera from clipping through walls
- Movement triggers the screen fade effect to reduce VR motion sickness

2. Core Variables

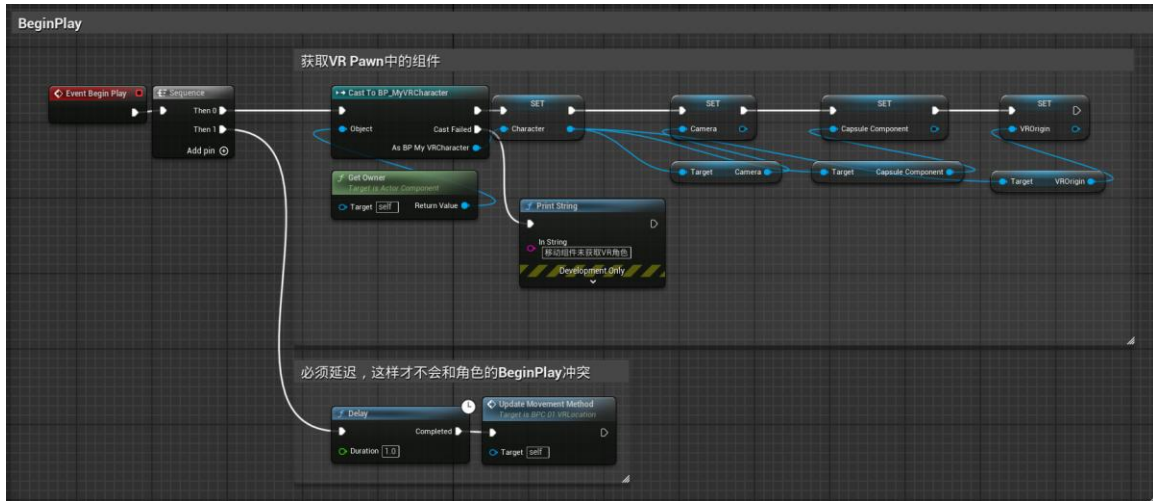
Variable	Type	Description
VRLocationMode	Enum	Current movement method enum value, controls input mapping switching
Input Mapping Context Array	Array of IMC	Array of input mapping contexts, one for each movement method
Move Speed	Float	Smooth locomotion speed multiplier
Can Player Move?	Boolean	Whether the player is allowed to move (can be disabled during gameplay)
Can Player Jump?	Boolean	Whether the player is allowed to jump

3. Initialization (BeginPlay)

Description

During initialization the component obtains references to the key components on the VR character (Character, Camera, Capsule, VR Origin), then calls UpdateMovementMethod with a delay to set the initial movement method.

Blueprint screenshots



Notes

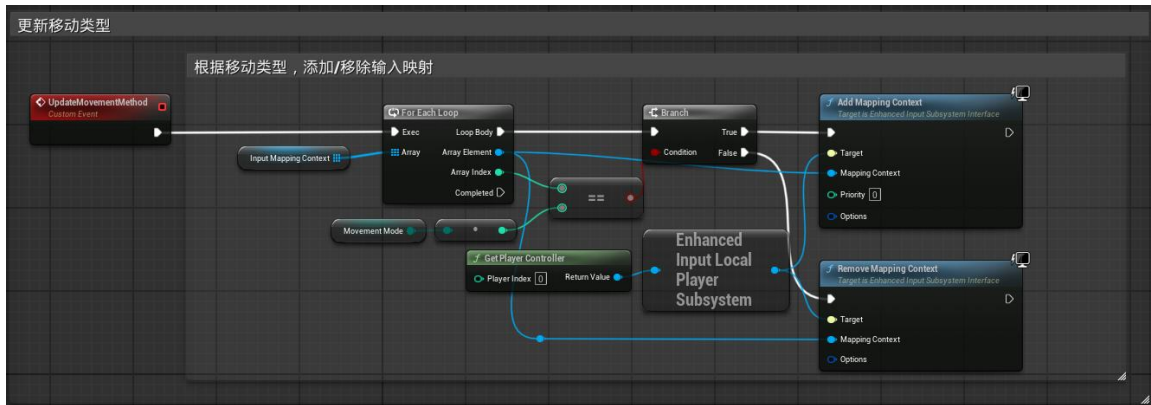
- UpdateMovementMethod must be executed with a delay, otherwise it conflicts with the character's BeginPlay initialization order
- If the Cast fails a message is printed; the developer should check that the component is correctly attached to the VR character

4. Switching Movement Methods (UpdateMovementMethod)

Description

Based on the VRLocationMode enum value, input mapping contexts are dynamically added and removed to switch the movement method.

Blueprint screenshots



Key points

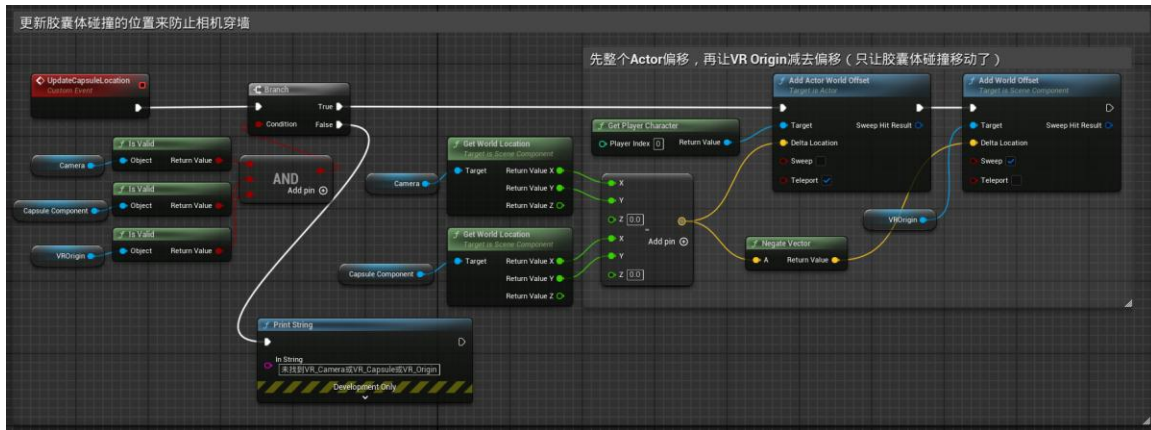
- Every time UpdateMovementMethod is called, all mappings are re-evaluated so that only the current movement method's mapping stays active
- To switch the movement method, simply change the MovementMode enum value and call UpdateMovementMethod

5. Updating the Capsule Position (UpdateCapsuleLocation)

Description

While the player moves, the capsule collision position is synchronized with the VR camera's horizontal position to prevent the camera from clipping through walls. This is a key safety mechanism in VR smooth locomotion.

Blueprint screenshots



Core principle

This uses an "offset first, then compensate in reverse" technique:

- The camera's local position is determined in real time by the VR headset (HMD) tracking; the code does not directly control the camera position
- But the camera's world position = VROrigin world position + HMD local offset, and VROrigin is the camera's parent
- Step 1: Move the whole Actor. VROrigin, being a child component, moves along with it. Without compensation the camera's world position would shift and the player would feel a jump in the view
- Step 2: Offset VR Origin in the reverse direction by the same distance, canceling the Actor's movement effect on VROrigin, bringing VROrigin back to its original position so the camera's world position stays unchanged
- Final result: only the capsule collision moves to the camera's horizontal position; VROrigin and the camera stay put, so the player feels no view jump

(IA_MoveForward)

Handles the thumbstick forward/back input, performing smooth locomotion along the VR camera's forward direction, and triggers the screen fade during movement to prevent motion sickness.

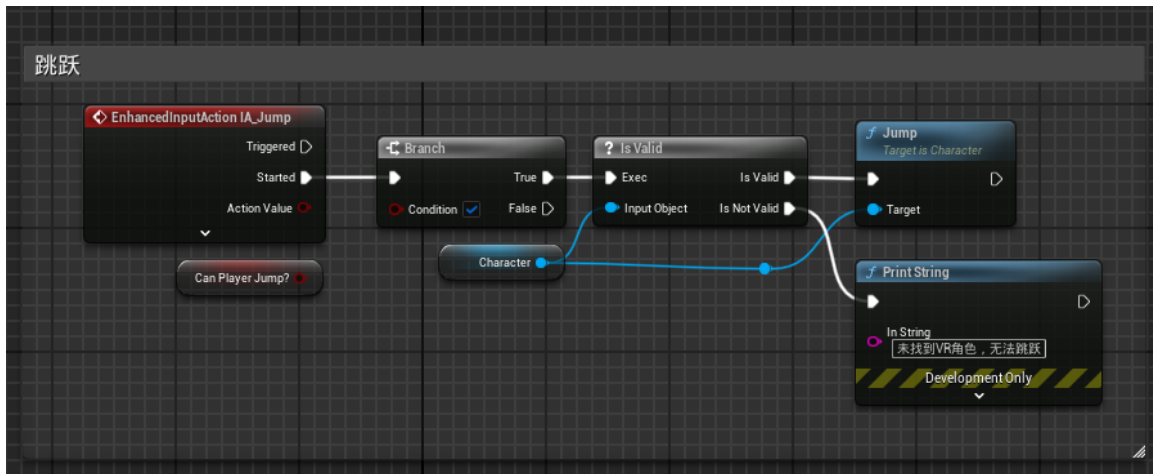
[illegible]

- The 0.01 dead zone threshold filters out thumbstick drift, preventing tiny inputs from causing unintended movement
- The camera's Forward Vector is used as the movement direction so movement matches the player's view
- The capsule position is updated before moving to guarantee accurate collision detection

(IA MoveHorizontal)

Description

Blueprint screenshots



Key points

- Started is used instead of Triggered so the jump fires only once instead of continuously